

Edit-Level Tracking of Narrative Changes in 10-K Filings

Xiao Li
School of Computer Science
University College Dublin
Dublin, Ireland
xiao.li@ucdconnect.ie

Changhong Jin
School of Computer Science
University College Dublin
Dublin, Ireland
changhong.jin@ucd.ie

Yingjie Niu
School of Computer Science
University College Dublin
Dublin, Ireland
yingjie.niu@ucdconnect.ie

Jingyi Wang
College of Computer Science
Beijing University of Technology
Beijing, China
jingyiwang@emails.bjut.edu.cn

Ruihai Dong
School of Computer Science
University College Dublin
Dublin, Ireland
ruihai.dong@ucd.ie

Abstract

The Form 10-K provides a detailed overview of a company's operations, strategies, competitive environment, and future outlook. Investors meticulously analyse annual changes in these reports to identify subtle shifts in disclosed information. However, financial narratives often contain extensive boilerplate text, which dilutes the efficacy of traditional sentiment analysis. We propose a novel edit-level analysis framework to isolate meaningful changes in Management Discussion and Analysis (MD&A) sections of 10-K filings. Using sentence embeddings, we classify each sentence in the filings for years t and $t + 1$ as Kept (repeated boilerplate), Deleted (removed information), and Added (newly introduced information). We then construct a net sentiment change signal from the difference in average FinBERT-tone sentiment between Added and Deleted content. This Core Sentiment measure has been demonstrated to outperform conventional bag-of-words sentiment measures. Furthermore, based on cleaned and standardised 10-K corpora (e.g., EDGAR-CORPUS) and our Flex_10K pre-processing pipeline, our approach provides a reproducible basis for quantifying changes between annual reports.

CCS Concepts

• Information systems → Information extraction; Sentiment analysis.

Keywords

Financial NLP, SEC 10-K filings, Text analysis, Sentence alignment

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1 Introduction

Publicly accessible web repositories, such as the SEC's EDGAR system, contain extensive collections of financial filings, offering a valuable opportunity for large-scale text mining in financial markets. Annual 10-K reports, in particular, are important disclosures that serve as "blueprints" of a company's progress and challenges [16]. Within each 10-K, the Management Discussion and Analysis (MD&A) section provides a rich narrative from the perspective of company managers, who discuss financial results, risks, and forward-looking plans. Financial analysts and investors are increasingly using textual analysis of MD&A to gain insights beyond the numerical data [5, 11].

However, the majority of the text remains unchanged from previous years. Approximately 76% of MD&A sentences are repeated from the previous year and contribute little new information [4], which obscures the value of new or revised statements [9]. Boilerplate contamination hinders the informative power of MD&A, and has resulted in two key limitations:

- (1) Dilution effect issue: the noise of stale, outdated, and uninformative 'platitudes' drowns out the actual changed signals, rendering traditional full-text sentiment scores exceedingly faint and ineffective.
- (2) Single score issue: Item-level sentiment scores reduce rich narratives to a single, coarse numerical value. Such scores are unable to track dynamic shifts in sentiment within the MD&A section.

In this study, our approach involves posing two key questions to address these gaps: firstly, "How did the text change?" and secondly, "Which parts of the text changed?" Our approach involves performing a semantic alignment of consecutive MD&A versions at the sentence level. By encoding each sentence with a sentence embedding model and employing bidirectional nearest-neighbour matching, it becomes possible to identify which sentences in the new report correspond to which sentences in last year's report, as well as those that have no counterpart. This process enables a precise classification of each sentence as either Kept (unchanged boilerplate), Deleted (present last year but removed now) or Added (newly added in the current year). Leveraging this granular decomposition, a novel Core Sentiment measure is constructed, defined as the difference in sentiment between the Added and Deleted

content. Intuitively, this measure captures the net change in management's tone: are the new disclosures this year more positive or negative relative to the information that was dropped or de-emphasised since last year? In summary, as regulatory filings and corporate disclosures are made available online in ever-growing volume, investors face an information overload analogous to other web content streams. Our study demonstrates the efficacy of using web-scale text data combined with financial natural language processing (FinNLP) algorithms to enhance information extraction and decision-making processes.

2 Related Work

2.1 Financial Text Analysis and Sentiment

In the early stages of textual analysis in the field of finance, simple dictionary methods had been applied to measure tone and sentiment in company disclosures and news reports. Pioneering work by Tetlock [15] showed that negative words in news media can predict stock price movements, spurring interest in text as a quantifiable data source. In the context of annual reports, researchers have observed that off-the-shelf sentiment lexicons are not well-suited to financial jargon. A domain-specific sentiment dictionary tailored for SEC filings has been introduced [12], which identifies words that carry positive or negative connotations in finance. Subsequent studies [6, 8] have demonstrated that the tone of MD&A can serve as a modest predictor of returns and risk, with firms employing more negative language tending to perform worse or face higher costs of capital. However, dictionary-based approaches have limitations in that the context and negation are ignored, and each word is treated independently, resulting in the potential misclassification of nuanced sentences. Recent advances have employed pre-trained language models to address these issues. FinBERT [1] is a BERT-based model that has been further trained on financial corpora, achieving higher accuracy in financial sentiment classification than generic BERT or dictionary methods.

2.2 Analysing Changes in Financial Disclosures

Brown and Tucker [2] propose a content modification score based on word-level changes and demonstrate that investors respond to MD&A alterations (stock returns around the filing date are more pronounced when the MD&A has changed significantly). Cohen et al. [3] introduced an approach that involved the assessment of the similarity between a firm's filing and its previous year's version. This was achieved by applying the similarity metric to TF-IDF vectors. The methodological framework enabled the classification of firms into two distinct categories: "changers" and "non-changers". Furthermore, Kim et al. [7] have identified a correlation between 10-K textual changes and tail risks, such as stock price crashes. While the significance of disclosure changes is acknowledged in these works, their methods address the document in its entirety or employ simplified metrics (e.g., word counts, similarity scores). Our study seeks to address this gap by introducing a sentence-level alignment approach inspired by techniques in time-series data mining (the concept of matrix profiles [17]).

Firm 10-K

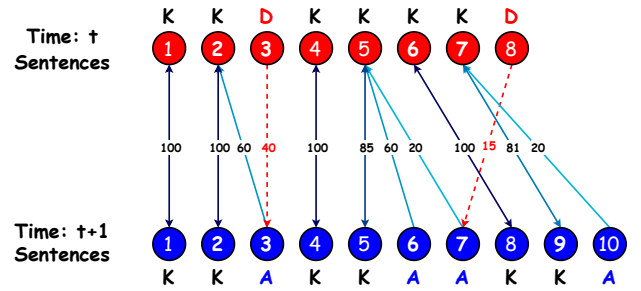


Figure 1: Bidirectional Alignment Between t and $t + 1$ 10-K Reports

3 Methodology

3.1 Data Collection

Our empirical analysis is based on annual 10-K filings for S&P 500 firms over the period 2005–2024. We use two structured, pre-cleaned corpora: EDGAR-CORPUS [13] (2005–2020) and our Flex_10K [10] (2021–2024), from which we directly obtain the Management Discussion and Analysis (MD&A, Item 7) text. We then link each filing to monthly stock returns from Yahoo Finance¹ and obtain the risk-free rate from the Kenneth R. French Data Library² in order to compute excess returns for subsequent portfolio tests.

3.2 Bidirectional Alignment and Sentence-level Similarity

We split each Item 7 section into sentences using a rule-based splitter, and embed every sentence with Sentence-BERT (SBERT, *all-mpnet-base-v2*). For each firm and adjacent year pair $(t, t + 1)$, we compute cosine similarities between all sentences in year t and all sentences in year $t + 1$, and for every sentence in year t , we store the index and similarity of its most similar counterpart in year $t + 1$. We also perform the reverse matching from year $t + 1$ to year t . This bidirectional matching gives each sentence in both years a single best counterpart and an associated maximum similarity, which we later use to infer edit types (See Figure 1).

3.3 KDA Edit Sets and Sentiment Aggregation

Using the bidirectional similarities from Section 3.2, we classify each sentence into three edit sets based on a cosine threshold τ :

- **Kept (K):** A pair of sentences (e_t, e_{t+1}) is marked as kept if they are mutual best matches across years and both similarities exceed τ .
- **Deleted (D):** Sentences in year t that do not belong to any kept pair and have no match above τ in year $t + 1$ are classified as deleted.
- **Added (A):** Sentences in year $t + 1$ that do not belong to any kept pair and have no match above τ in year t are classified as added.

¹Yahoo Finance (via yfinance): <https://ranaroussi.github.io/yfinance/>

²https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

For the threshold choice, prior work with SBERT typically calibrates effective cosine thresholds in the 0.7–0.8 range [14]. Starting from this guidance, our in-house experts inspected 10-K Item 7 sentence pairs and, given their highly templated language, recommended a slightly stricter cutoff. We therefore set $\tau = 0.8$ in the baseline and confirm robustness to nearby values ($\tau \in [0.75, 0.85]$).

For sentence-level sentiment, we use FinBERT-tone [1], a FinBERT variant specifically trained for financial sentiment analysis. For each sentence e , FinBERT-tone yields class probabilities $P(pos)$, $P(neg)$, $P(neu)$. We map these to Equation 1, thus obtaining a net sentiment score $s(e)$ for every sentence.

$$s(e) = P(pos) - P(neg) \in [-1, 1] \quad (1)$$

For firm i and year $(t, t+1)$, we compute $\mu_K(i, t)$, $\mu_D(i, t)$, $\mu_A(i, t)$ as the mean sentiment within the Kept, Deleted, and Added sets, respectively. Our edit-level signal is the net edit tone (KDA Score):

$$KDA_{i,t+1} = \mu_A(i, t) - \mu_D(i, t) \quad (2)$$

which is high when the added text is more positive than the deleted text and low in the opposite case. As a document-level baseline, we also compute Item 7 sentiment at year $t+1$ by averaging $s(e)$ over all sentences without distinguishing edit types.

3.4 Portfolio Construction and Backtesting

To evaluate whether tracking sentence-level edits can extract a more informative sentiment signal from standard document-level Item 7 tone, we conduct a simple long-short backtest. At the beginning of each calendar month, we assign to each stock two signals derived from its most recent 10-K filing: (i) the document-level Item 7 sentiment at year $t+1$, obtained by averaging $s(e)$ over all Item 7 sentences and used as our baseline text-based signal, and (ii) the edit-level KDA sentiment score $KDA_{i,t+1}$ constructed from the corresponding Item 7 pair.

Excess returns and cumulative long-short returns are defined as follows. For stock i in month m , the excess return is:

$$r_{i,m}^{\text{excess}} = r_{i,m} - r_m^f \quad (3)$$

where $r_{i,m}$ is the stock return and r_m^f is the risk-free rate.

For a given month m , we sort stocks into quintiles based on a given signal (document-level or KDA). Let $Q_{k,m}$ denote the set of stocks in quintile $k \in 1, \dots, 5$ in month m . The equal-weighted excess return of quintile k is:

$$r_{Q_{k,m}}^{\text{ew}} = \frac{1}{|Q_{k,m}|} \sum_{i \in Q_{k,m}} r_{i,m}^{\text{excess}}. \quad (4)$$

The corresponding long-short portfolio return is:

$$L_m = r_{Q_{5,m}}^{\text{ew}} - r_{Q_{1,m}}^{\text{ew}}, \quad (5)$$

and the cumulative long-short return over a backtest horizon of T months is:

$$R_{\text{cum}} = \prod_{m=1}^T (1 + L_m) - 1. \quad (6)$$

For each signal separately, we sort stocks into quintiles based on their cross-sectional signal value. We form equal-weighted portfolios for each quintile and construct a long-short factor that buys the most positive quintile (Q_5) and shorts the most negative quintile (Q_1). Portfolios are rebalanced at the end of each month and held for one month. Because 10-K filings arrive throughout the year and firm-level signals only update when a new filing is observed, positions naturally overlap across time. Our hypothesis is that, by focusing on added and deleted sentences rather than averaging over the entire section, the KDA sentiment score yields stronger and more persistent positive quintile return differentials than standard Item 7 sentiment.

4 Results

4.1 KDA Results

Table 1 illustrates the KDA decomposition using Apple’s 2023/2024 Item 7 disclosures. The Kept block consists of almost verbatim legal boilerplate about ongoing proceedings, which contributes little to year-on-year variation in sentiment. By contrast, the added sentence in 2024 introduces “Apple Intelligence™... that uses generative models”, while the deleted sentence removes a 2023 focus on the launch of “Apple Vision Pro™... expected to be available in early calendar year 2024”. This example shows that KDA isolates narrative updates, specifically what firms newly highlight or quietly drop, rather than boilerplate, which provides a more targeted sentiment signal for the portfolio tests that follow.

Table 1: Examples of Kept, Deleted and Added sentences between Apple 2023/2024 Item 7

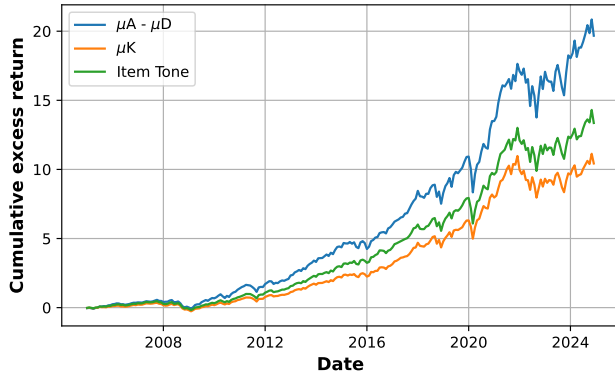
Type	Year	Sentence
Kept	2023	The Company is subject to various legal proceedings and claims that arise in the ordinary course of business, the outcomes of which are inherently uncertain.
	2024	The Company is subject to various legal proceedings and claims that arise in the ordinary course of business, the outcomes of which are inherently uncertain.
Added	2024	Apple Intelligence™, a personal intelligence system that uses generative models.
Deleted	2023	Apple Vision Pro™, the Company’s first spatial computer featuring its new visionOS™, expected to be available in early calendar year 2024.

4.2 Backtest Results

Figure 2 shows that all three sentiment-based strategies generate steadily rising cumulative excess returns, indicating that KDA scores contain tradable information. The portfolio based on $(\mu_A - \mu_D)$ consistently dominates, especially after 2010, while Item Tone

Table 2: Portfolio performance of sentiment-sorted quintiles (Q5: most positive; Q1: most negative).

Type	Sharpe Ratio		Max Drawdown		Annualised Return		Annualised Volatility	
	Q5	Q1	Q5	Q1	Q5	Q1	Q5	Q1
Item Tone	1.03	0.71	-42.57%	-54.62%	16.17%	13.77%	15.63%	19.37%
μ_K	0.95	0.80	-44.53%	-50.73%	15.05%	15.01%	15.87%	18.83%
$\mu_A - \mu_D$	1.06	0.70	-39.93%	-58.05%	18.27%	12.60%	17.25%	18.13%

**Figure 2: Ablation of KDA score aggregation in backtests: cumulative portfolio excess return.**

lies in the middle and (μ_K) is weakest. This suggests that explicitly contrasting added and deleted disclosures yields the strongest return signal.

Table 2 confirms this pattern in risk–return statistics. The ($\mu_A - \mu_D$) strategy delivers the highest Q5 annualised return and Sharpe ratio, with a relatively contained max drawdown compared with the other variants. At the same time, its Q1 portfolio earns noticeably lower returns, implying a clearer performance spread between “most positive” and “most negative” firms.

5 Conclusion

The approach adopted has been to focus on changes in annual reports rather than simply on the presence of such changes. The limitations of extant approaches in terms of boilerplate dilution and single-score analysis have been a source of motivation for the introduction of an edit-level semantic analysis framework for MD&A sections of 10-K filings. By aligning consecutive reports at the sentence level using Sentence-BERT embeddings, the present study decomposed each filing into Kept, Deleted, and Added content. This decomposition was then used to define a Core Sentiment measure based on the net tone of Added versus Deleted text.

The empirical evidence obtained has demonstrated that this Core Sentiment measure is both statistically and economically significant. In the context of portfolio backtests, strategic approaches that involve investing in firms with strongly positive Core Sentiment have been shown to generate substantial risk-adjusted returns and significantly higher Sharpe ratios in comparison to baselines. The findings of this study indicate that markets exhibit a tendency to underreact not only to the presence of textual change but also to

the sentiment embedded in the new information that managers elect to add or delete.

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